

# Student Management System:

## Project Proposal

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### 1. Project Overview

The Student Management System is a C++ program used by schools for managing records of students and teachers. Programming that is Object-Oriented stores and organizes data simply.

#### Main Purpose:

- Add and manage student records
  - Add and manage teacher information
  - Search and view all records
  - Learn OOP concepts practically
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### 2. System Features

#### Student Management:

- Add students with name, age, roll number, course, and marks
- Automatic grade calculation (A+, A, B, C, F)
- Search students by roll number
- View all student records

#### Teacher Management:

- Add teachers with name, age, subject, and experience
- View all teacher records

#### Interface:

- Simple menu-driven system
  - Easy navigation
  - Console-based interface
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### 3. Technical Details

**Programming:** C++ with Object-Oriented Programming **Storage:** Arrays (50 students, 20 teachers maximum) **Classes Used:**

- Person (base class)
- Student (inherits from Person)
- Teacher (inherits from Person)
- StudentManagementSystem (main controller)

#### OOP Concepts:

- Inheritance
  - Polymorphism
  - Virtual functions
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### 4. How It Works

#### Menu Options:

1. Add Student
2. Add Teacher
3. Show All Students
4. Show All Teachers
5. Search Student
6. Polymorphism Demo
7. Exit

**Usage:** Simply run the program and choose menu options to perform different tasks.

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### 5. Benefits

#### Educational:

- Learn OOP programming practically
- Understand inheritance and polymorphism
- Practice with classes and objects

#### Practical:

- Easy record management for small schools
  - Quick student information access
  - Organized data storage
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## 6. Implementation Strategy

### Development Approach:

- Start with base Person class design
- Implement Student and Teacher classes using inheritance
- Create management system with array-based storage
- Add menu-driven user interface
- Test each function individually

### Implementation Steps:

1. Design class structure
  2. Code basic classes (Person, Student, Teacher)
  3. Implement management system functions
  4. Create user interface menu
  5. Testing and debugging
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## 7. Conclusion

Practically this Student Management System shows Object-Oriented Programming concepts. It successfully implements inheritance, along with polymorphism, including virtual functions. It also functions in a useful way that manages educational records.

Target Users: Programming students, small schools Learning Value: It is outstanding in understanding OOP fundamentals